FROM OUR FELLOWS JUST ENVIRONMENTS URBANISM AND INFRASTRUCTURE

The Banality of Infrastructure

by Nikhil Anand

June 27, 2017 (June 27, 2017)

Three years ago, in April 2014, the city of Flint, Michigan, switched its water source from the Detroit Water and Sewerage Department to the Flint River. This, in many ways, was an unremarkable decision, made in part to save monies while a new pipeline was being built. The move was expected to save the cash-strapped city of Flint $5 million a year. Between September 2014 and January 2015, residents of Flint began to notice something amiss with their water. It smelled and had a strange color. Others began to complain of skin rashes, of hair loss, and sickness. City officials waved away their concerns, while at the same time dousing the supply with larger quantities of chlorine, so as to disinfect it. By the time a pediatric study noted elevated levels of lead in Flint's children in September 2015, residents had been exposed to contaminated supplies for over a year. Over six thousand children were documented as having elevated levels of lead in their blood. By the end of the year, the city of Flint declared a state of emergency. Civil and criminal lawsuits have since been filed against key government officials. Earlier this year, the EPA awarded Flint $100 million for infrastructure upgrades, and the state government settled a lawsuit, agreeing to pay an additional $97 million toward replacing Flint's lead and galvanized iron water lines. Yet most experts now say the problem far exceeds the nearly $200 million that have been allocated for its repair.

In my first book, *Hydraulic City*, I explore how water services are a key site for the making of urban inequality and inclusion in Mumbai.[1] In their everyday access to urban water supply, residents in Mumbai recognized important processes with which they were seen and treated as deserving (or undeserving) citizens of the city. They identified water services as a vital, materially instantiated form of the social contract between the state and the citizen, and questioned the morality of government absent a fundamental and basic effort to provision water. In different settlements of Mumbai, residents would frequently ask: what legitimacy does the state have, if it is not providing them with safe and dependable supplies of water? Yet, as the events in Flint (as in Mumbai) show so vividly, water infrastructures are not just the location for the making of urban citizens. They are also ongoing processes through which the social contract is violently betrayed or denied, where residents are either cut off or poisoned by the vital services they need to live.[2]

As the liberal social contract frays at the seams not just in cities of the global South but also in the United States, liberal modes of accounting and accountability have been unable to deal with crises proliferating in and around different infrastructures. As bridges fall into the Mississippi, levees breach in New Orleans, and power failures and rail accidents plague the eastern seaboard of the United States, events of infrastructure disrepair and breakdown seem to be everyday conditions of the political present. These rather spectacular events are not only sites for performances of environmental injustice. They also reveal how infrastructures operate with multiple temporalities that distribute life and harm. As they accrete and distribute resources over time, infrastructures call for an accounting of the world that recognizes how past histories of injustice are remade and realized anew.

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For a while, the news reporting around Flint generated outrage in the United States. For many, it represented one more instance of the environmental injustice to which communities of color are subject. Critics correctly pointed out that the provision of safe water would not be subject to the same stringent controls of cost savings had Flint been wealthier and white. The histories of public service provision and environmental protection in American cities substantiate this critique.[3] The water crisis in Flint also reveals, in vivid detail, how the violence of environmental injustice is generated in the most banal of ways—through bureaucratic routines of boredom and calculation that do not account for the humans they affect.[4] In Flint, environmental harms proliferated when state officials made banal decisions to read, direct, and evaluate the safety of the toxic material infrastructures they managed.

The banality of infrastructure

In her agenda-setting essay, "The Ethnography of Infrastructure," Susan Leigh Star emphasizes the importance of studying the dull materialities of infrastructure. "Study a city and neglect its sewers and power supplies," she writes, "and you miss essential aspects of distributional justice and planning power."[5] Indeed, urban infrastructures have long escaped the attention long accorded to research in housing, public space, and social movements. As technological networks many learn not to see, the invisibility of infrastructure sometimes is described as a kind of revelation.

Infrastructures are not so much invisible, however, as they are "boring."[6] For many with social and political power, they are mundane and banal features of urban living.[7] The dullness of infrastructure has political effects. It enables their various managerial authorities—officials in public utility commissions and departments of environmental services—to remain faceless. It allows their practices to remain illegible in opaque institutions. As the crisis in Flint revealed, their "boringness" obscures how the work they are made to do is fundamentally political and vital to understanding how injustice and natural resources are distributed.

For over five decades, Flint depended on the Detroit Water and Sewerage Department to source and supply it with water. This relationship has been fraught, particularly in recent years, when the city of Detroit has had to lean on its water department for larger revenue contributions. In 2013, the city of Flint voted to leave the Detroit water department and join a new regional water authority. Yet, the new project triggered a contentious water dispute with Detroit. When the state approved Flint's request to leave Detroit water services in 2013, the Detroit water department cancelled Flint's water contract effective the following year, two years before the new water authority would become operational. Pushed against the wall, Flint city officials voted not to negotiate with Detroit, but to use treated water from the Flint River for a period of two years while the pipes of the new authority were being installed. Mobilizing their expert authority, managers insisted they could take the appropriate steps to ensure that water in the Flint River was made fit and safe for drinking.

Present histories

Humans are thrown into worlds not of their making,[8] into worlds already structured by infrastructures. While infrastructures are present, they also continue to reproduce the political

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relationships of the times in which they were constructed. In her wonderful book, *Democracy's Infrastructure*, Antina von Schnitzler has shown how in apartheid-era South Africa water infrastructures continue to reproduce their politics in the post-apartheid era.[9] The inequalities of the water network—materialized in pipe designs, meters, and valves—continue to structure relationships between the formerly segregated city of Johannesburg, and the townships around it. Stephen Collier describes this as the "intransigence" of infrastructure.[10] Network designs and technologies engineered in a particular political moment continue to reproduce the political worlds of the times in which they were engineered.

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Similarly, the crisis in Flint ensued because of administrative decisions taken both in the present and past. Flint's pipes were first installed to serve a rapidly growing city in the early-mid-twentieth century. Powered with cheap water, energy, and a booming auto industry, Flint's population grew rapidly in this period, from a hundred thousand in 1920 to nearly two hundred thousand in 1960. Its water network was expanded with lead service lines that were quick and easy to install. Yet the growth of the city was not sustained. Following a set of events familiar to other rustbelt cities, such as federal housing and loan programs that incentivized suburbanization, coupled with deindustrialization, these policies decimated the city over subsequent decades.[11] In her piece "Thinking with Flint," Malini Ranganathan describes carefully the ways in which these historical processes played a significant role in creating the conditions for the Flint water crisis.[12] Drawing on the work of Ruth Wilson Gilmore and Laura Pulido, Ranganathan argues that property abandonment in Flint in the late twentieth century made its infrastructural abandonment all but inevitable. Federal home loans, urban policies, and layoffs at General Motors nurtured a rapid depopulation of Flint, and generated a city that was Black and impoverished in the late twentieth century.

Read Malini Ranganathan's contribution to the "Just Environments" series here.

Political processes over the last five decades have produced both citizens and the city as fiscally insolvent and unworthy of credit, and unable to respond to an expansive crumbling infrastructure demanding maintenance and investment. The switch to cheaper water was necessitated therefore not just because of contemporary austerity politics, but also an effect of trying to support an expansive aging water system that serves far fewer citizens than what it had been designed for.

The will to ignore

Throughout the entire controversy, health and environmental regulators working for the Michigan Department of Environmental Quality (MDEQ) repeatedly insisted the water was safe, regularly downplaying and rejecting the efforts of parents, doctors, and scientists establishing otherwise. Next, when they recognized the water did have high levels of coliform bacteria (i.e., *E. coli*), they proceeded to treat it with high levels of chlorine, but not with corrosion-control phosphates that would protect the pipes from leaching lead. Nor did they think it necessary to test the water for lead for the first six months. Finally, when testing for lead in city water supplies, MDEQ officials adopted protocols that would studiously avoid finding lead. They selected homes that were less likely to have lead pipes, flushed the lead out of water pipes before they collected their samples, etc.

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Taken together, these activities demonstrate how the MDEQ worked hard *not* to find lead in the water—to remain ignorant of the problem. To the extent they conducted tests, they were directed toward ensuring that any questions or concerns around *question the work of the MDEQ in ensuring water safety."*

water safety could be dismissed as not based in science. In its review of the work of the MDEQ, the Flint Water Advisory Task Force, commissioned by the governor of Michigan, was scathing in its assessment of the ways in which the MDEQ sought to bury and subvert knowledge about dangerous levels of lead and bacteria in Flint's water. I quote here from their report:

> We believe that in the Office of Drinking Water and Municipal Assistance (ODWMA) at MDEQ, a culture exists in which "technical compliance" is considered sufficient to ensure safe drinking water in Michigan. This minimalist approach to regulatory and oversight responsibility is unacceptable and simply insufficient to the task of public protection… Throughout 2015, as the public raised concerns and as independent studies and testing were conducted and brought to the attention of MDEQ, the agency's response was often one of aggressive dismissal, belittlement, and attempts to discredit these efforts and the individuals involved.[13]

For the MDEQ conducting tests that established water safety was generative so long as the water was not found to be unsafe. Unsafe water produced problems, not just for public health, but also because finding water as such would bring into question the work of the MDEQ in ensuring water safety. Rather than risk this disruption, MDEQ officials worked to ignore the lead in water pipes. In finding water services compliant with standards, they sought to keep the structural violence of Flint's water infrastructure hidden underground. They worked hard to ensure water remained a dull and monotonous matter, one with a history and future that could be made boring enough that no one would notice and demand otherwise.[14]

Conclusion

In Flint, as in Mumbai, environmental injustice is produced through water pipes following a series of interactions between human and nonhuman actors. These interactions are not novel, nor are they merely reenactments of longer histories of structural (dare I say infrastructural) racism. Instead, environmental injustice emerges through old-new articulations of banal infrastructures that are always in the process of being made and unmade in environments of produced ignorance. Indeed the disaster cannot be understood without examining the relationships of race, inequality, and difference with which these infrastructures are made, and are maintained with infrastructures as vital and as toxic as lead pipes. Yet, the Flint water disaster ensued also because of a set of bureaucratic decisions to switch to the river as source, and modified water management protocols designed to find this water as "safe." Imbibed by children and adults alike, the lead and *E. coli* of Flint's water pipes produced old-new events of harm and suffering on political subjects that were overwhelmingly black and poor.

The Flint water crisis shows how environmental injustice is generated and administered through banal bureaucratic and technocratic infrastructures. Infrastructures are key to the making and distribution of both life and death. They are vital sites at which injustice is produced. To imagine and to make environments of justice, then, is necessarily to engage in the "boring" technopolitics of infrastructure; to reveal, refuse, and revolt against the ways in which their vital and violent politics are frequently obscured and buried from view.

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discrimination | global health | inequality | infrastructure | social class | water

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